

Microbiology: Hemorrhagic Viral Infections

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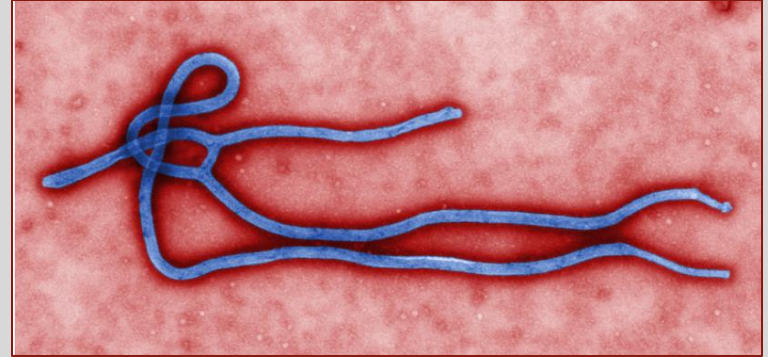


Session Objectives

1. Describe the characteristics of viruses that cause hemorrhagic fevers.
2. Compare and contrast the viral families involved in hemorrhagic fever infections.
3. Identify disease manifestations caused by hemorrhagic viral infections.
4. Summarize the epidemiology, transmission, pathogenesis, clinical features, diagnosis, management, and prevention of each of the hemorrhagic viruses presented.

Introduction to Hemorrhagic Viruses

- Families of viruses that cause **Viral Hemorrhagic Fever (VHF)**:
 - *Arenaviridae*
 - *Bunyaviridae*
 - *Filoviridae*
 - *Flaviviridae*
- Severe multisystem syndrome
- Vascular system is most affected leading to hemorrhage
- Symptoms range in severity from mild to life-threatening



Introduction to Hemorrhagic Viruses

- Characteristics of most viruses that cause VHF are:
 - Single-Stranded
 - RNA
 - Enveloped
- Natural reservoir: animals/environment
 - Humans are not a natural reservoir for these diseases
- Vectors

<u>Viral Family</u>	<u>Virus</u>	<u>Disease</u>	<u>Reservoir/Vect or</u>
Arenavirus	Lassa Virus	Lassa Fever	Reservoir - Rodents (Rats)
Bunyavirus	Hantavirus	Hemorrhagic Fever with Renal Syndrome	Reservoir - Rodents (Mice)
Flavivirus	Dengue Virus	Yellow Fever	Vector - Aedes Mosquito
Flavivirus	Yellow Fever Virus	Dengue Fever	Vector - Aedes Mosquito
Filovirus	Ebola Virus	Ebola Hemorrhagic Fever	Reservoir - Fruit bat or primates
Filovirus	Marburg Virus	Marburg Hemorrhagic Fever	Reservoir - Fruit bat or primates

Hemorrhagic Viruses

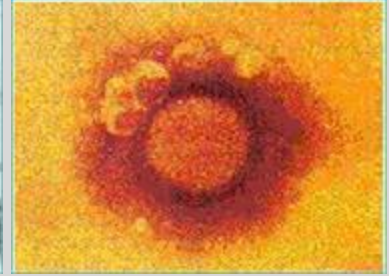
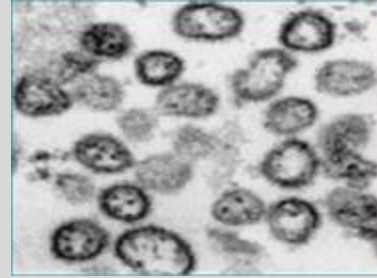
Viral Family	Virus	Disease
Arenaviruses	Lassa virus	Lassa Fever
	Junin Virus	Argentine Hemorrhagic Fever
	Machupo Virus	Bolivian Hemorrhagic Fever
	Sabiá Virus	Brazilian Hemorrhagic Fever
	Guanarito Virus	Venezuelan Hemorrhagic Fever
Bunyaviruses	Hantavirus	Hemorrhagic Fever with Renal Syndrome
	Nairo Virus	Crimean-Congo Hemorrhagic Fever
	Rift Valley Fever Virus	Rift Valley Fever
Flaviviruses	Dengue	Dengue Fever
	Yellow Fever Virus	Yellow Fever
Filoviruses	Ebola	Ebola Hemorrhagic Fever
	Marburg	Marburg Hemorrhagic Fever

Arenaviruses

- Enveloped, Spherical, -ss RNA, Segmented genome
- Rodents are natural reservoir
- Transmission from contaminated urine/feces from rodents
 - Direct contact: Consumption, broken skin
 - Inhalation of aerosolized particles
 - Person-to-person: Contact with blood or excretions from contaminated person
- Incubation period averages 10 -14 days
- Infect macrophages, induce interferon release → promote cell and vascular damage
- Cell-induced immunopathologic effects significantly exacerbate tissue destruction
- Lassa Virus, Machupo Virus, Junin Virus, Sabiá Virus, Guanarito Virus

Lassa Virus

- Family Arenaviridae
- Location: West Africa (Guinea, Nigeria, Liberia, Sierra Leone)
- Transmission: “multimammate rat” (*Mastomys natalensis*) – infected urine and feces



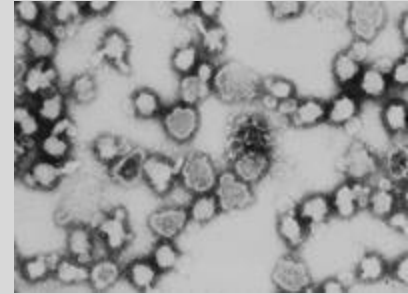
Lassa Virus

- Clinical features: fever, headache, muscle aches, malaise, sore throat, cough, n/v/d, chest pain, abdominal pain, hemorrhage. If severe: organ failure and shock
- 80% cases are mild (20% severe)
- ELISA/RT-PCR
- Ribavirin & supportive
- Most common complication from Lassa Virus = deafness (~1/3 of cases some degree of hearing loss)



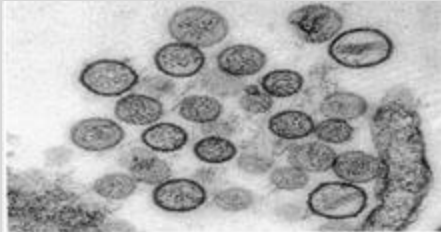
Bunyaviruses

- Enveloped, Spherical, -ss RNA, Segmented Genome
- Transmission:
 1. Most are arboviruses – transmitted by arthropods – mosquitoes, ticks, flies
 2. Hantavirus - aerosols from rodent urine and feces
- Initial viremia – flu-like illness
- Secondary viremia - virus may target specific tissues some causing encephalitis, hepatic necrosis, or hemorrhagic disease
- Hantavirus, Rift Valley Fever Virus, Nairo Virus

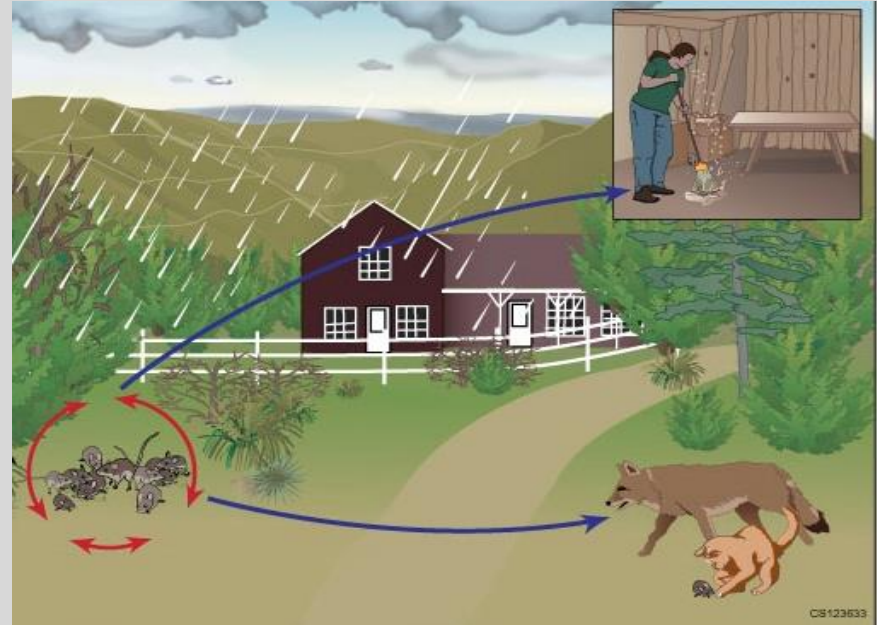


Hantavirus

Transmission: Rodents –
inhalation of aerosolized
virus from urine, droppings
Broken skin, mucous
membranes

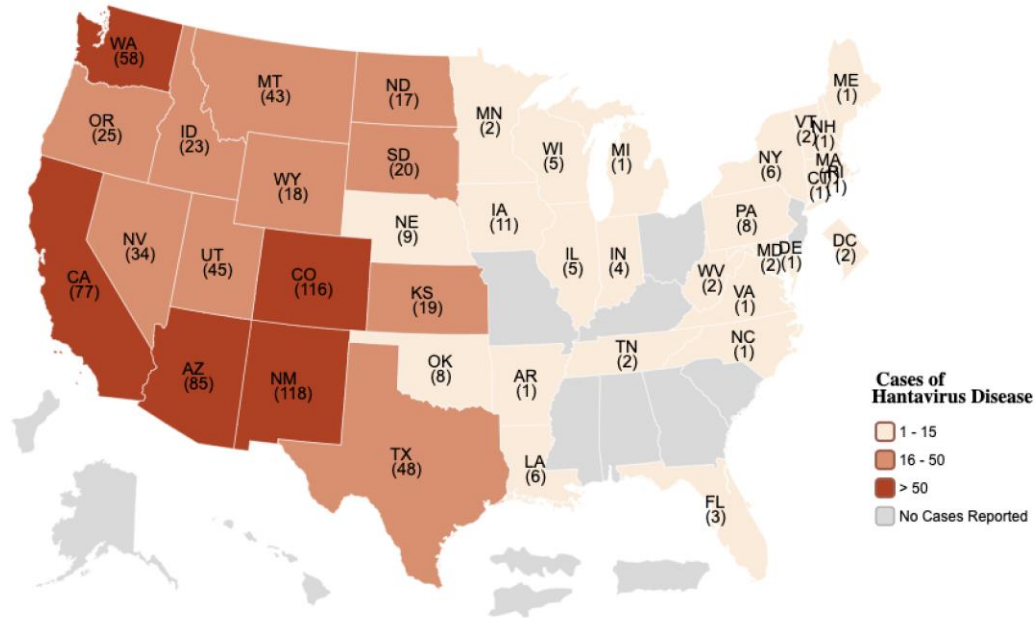


Source: Gray MR, Speath KB. The Bacteriology Sourcebook.
<http://www.elsevier.com>
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Hantavirus

Map of US Cumulative Cases of Hantavirus by State through 2020



All cases were confirmed between 1993-2020 and met the NNDSS case definition applicable at the time of reporting. Included in the sum total are 31 historical cases that occurred prior to 1993, but were confirmed retrospectively. Five cases had presumed exposure outside the United States.

Hantavirus

- Symptoms develop 1-2 weeks after exposure
- Initially: Fever, chills, headache, nausea, abdominal pain, back pain, redness of eyes, blurred vision
- May progress to hypotension, acute shock, vascular leakage and acute kidney failure
- Hemorrhagic Fever with Renal Syndrome
- Treatment is mainly supportive
- Prognosis varies depending on serotype (mortality 1-15%)
- Prevention: rodent control -primary strategy. Avoid disturbing rodent urine, droppings, saliva and nesting materials



Patient with Hemorrhagic Fever with Renal Syndrome displaying conjunctival hemorrhage

Flaviviruses

- Enveloped, (+) ssRNA
- Transmission
 - Female mosquitoes from infected vertebrate host to human
- Virions mainly named after geographic location
- Dengue Fever Virus, Yellow Fever Virus, St. Louis encephalitis, West Nile Virus, Zika, Hepatitis C

Pathogenesis of Flaviviruses

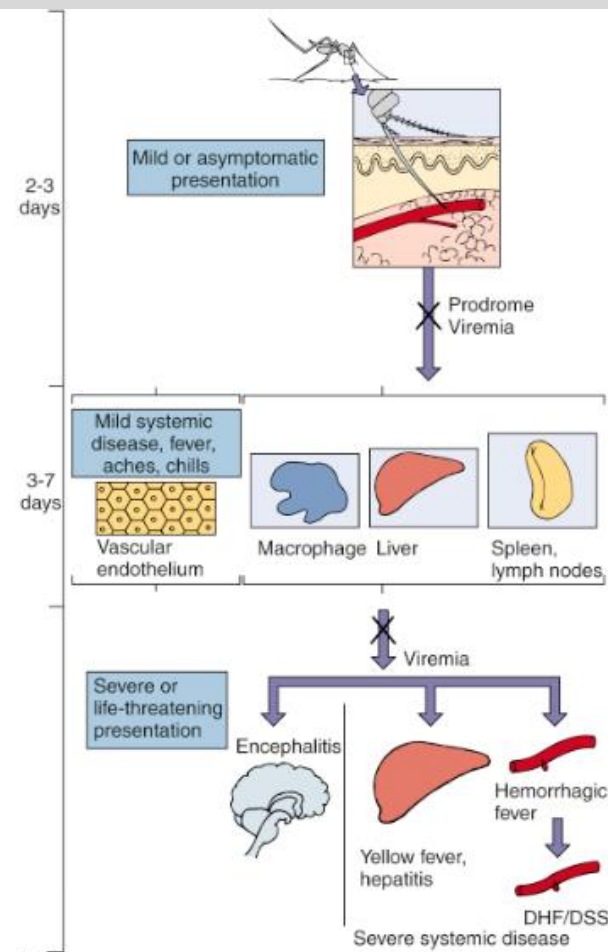


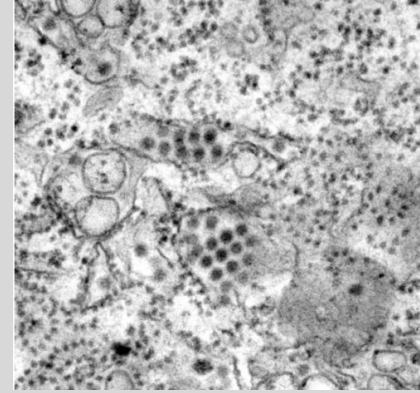
Fig. 52.4

Disease syndromes of the alphaviruses and flaviviruses. Primary viremia may be associated with mild systemic disease. Most infections are limited to this. If sufficient virus is produced during the secondary viremia to reach critical target tissues, then severe systemic disease or encephalitis may result. If antibody is present (X), viremia is blocked. For dengue virus, rechallenge with another strain can result in severe dengue hemorrhagic fever (DHF), which can cause dengue shock syndrome (DSS) because of the loss of fluids from the vasculature.

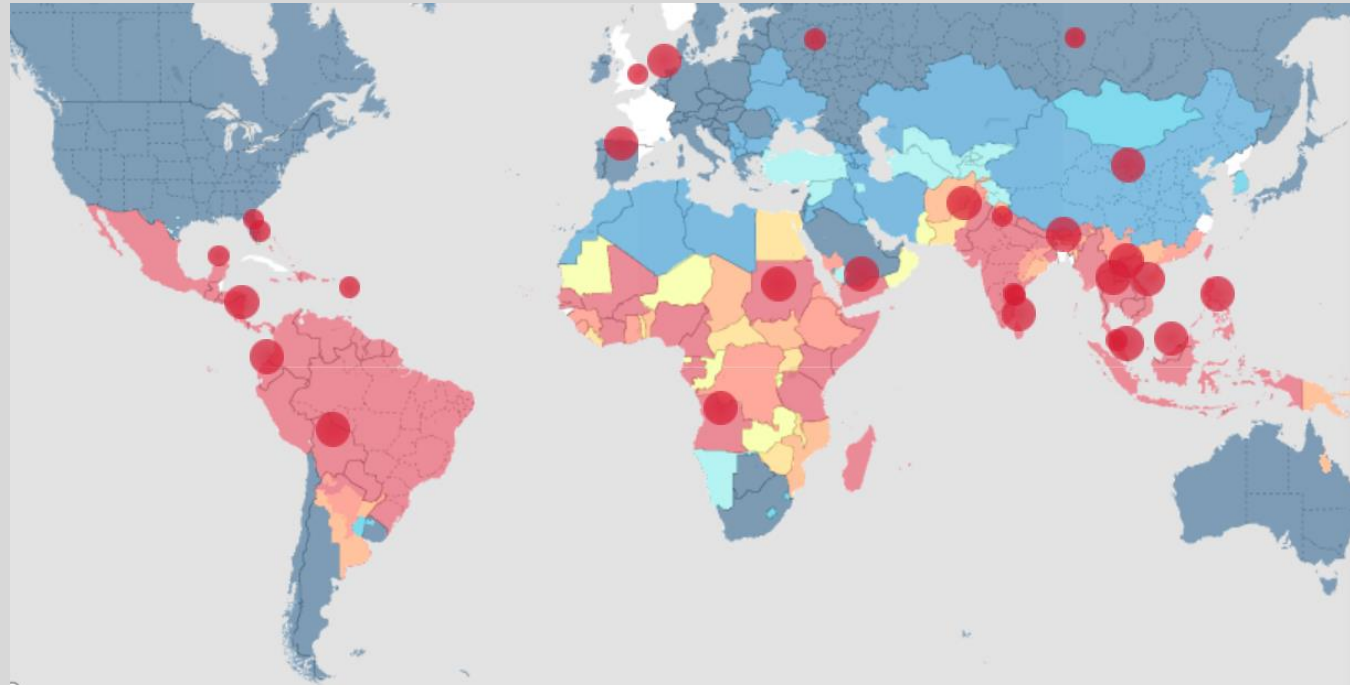
Murray PR, et al.,
Medical Microbiology,
9th edition. 52, 515-
526.e1

Dengue Fever Virus

- Flavivirus
- Southeast Asia, Western Pacific, Latin America and Africa
- Transmission by *Aedes aegypti* Mosquito



Dengue Fever Virus



HealthMap Reports

Recent reports of local or imported dengue cases from official, newspaper, and other media sources.

Source.

● Country Level ● Local Level

Global Consensus Map (2013)

Risk areas determined by consensus between sources including: national surveillance systems, published literature, questionnaires and formal and informal news reports. Source.

Absent Unlikely Uncertain Likely Present

CDC Yellow Book Map (2012)

CDC Yellow Book Map (2010)

Endemic risk areas determined using data from Ministries of Health, international health organizations, journals, and knowledgeable experts.

Source.

■ Endemic Area

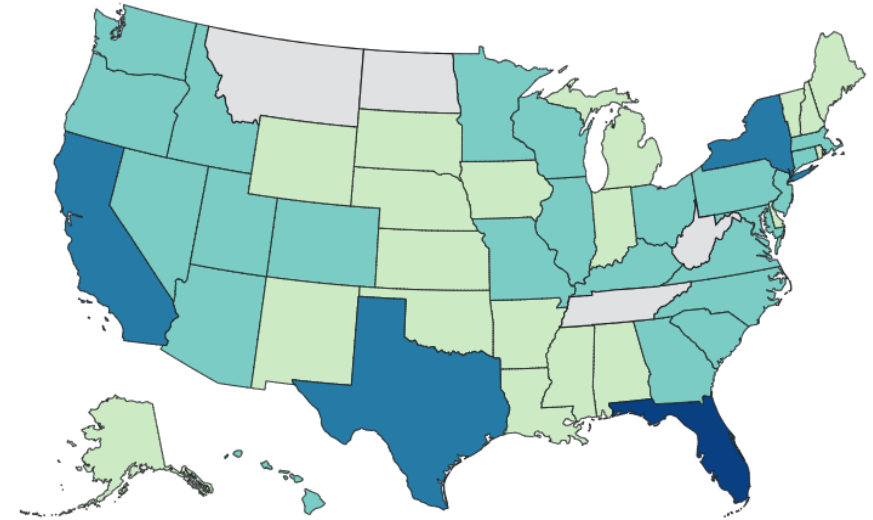
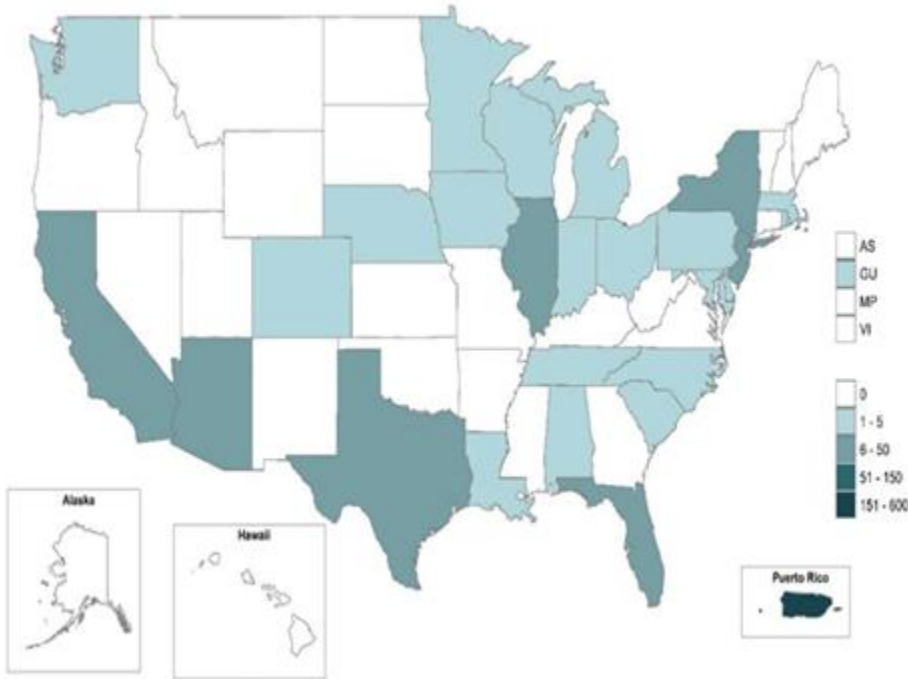
Dengue Fever Virus

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States and territories reporting dengue cases – United States, 2021 (as of January 5, 2022)

All dengue cases by jurisdiction of residence in US states and territories, 2025



U.S. territories

Freely associated states

AS GU MP PR VI FM MH PW

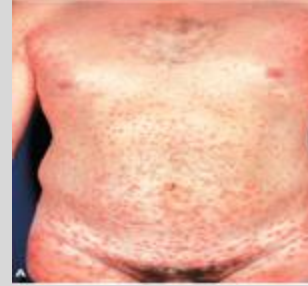
Legend

0 1 to 4 5 to 49 50 to 249 250+

*CDC reports provisional dengue case counts reported to ArboNET for the United States and its territories on the first Thursday of each month.

Dengue Fever Virus

- Incubation period: 4-7 days
- Fever, headache, rash, Myalgia, arthralgia (Break-bone fever), Thrombocytopenia, Retro-orbital pain, Septic Shock
- Dengue Hemorrhagic Fever
 - Potentially deadly complication of dengue fever
 - High fever
 - Hemorrhagic (often with hepatomegaly)
 - In severe cases, circulatory failure may occur
 - Most serious in children
 - bleeding occurs 3-5 days after fever
- Similar presentation to Chikungunya infection

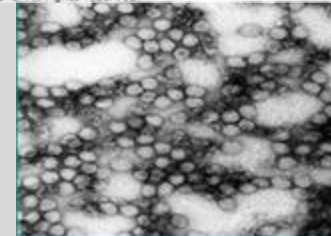
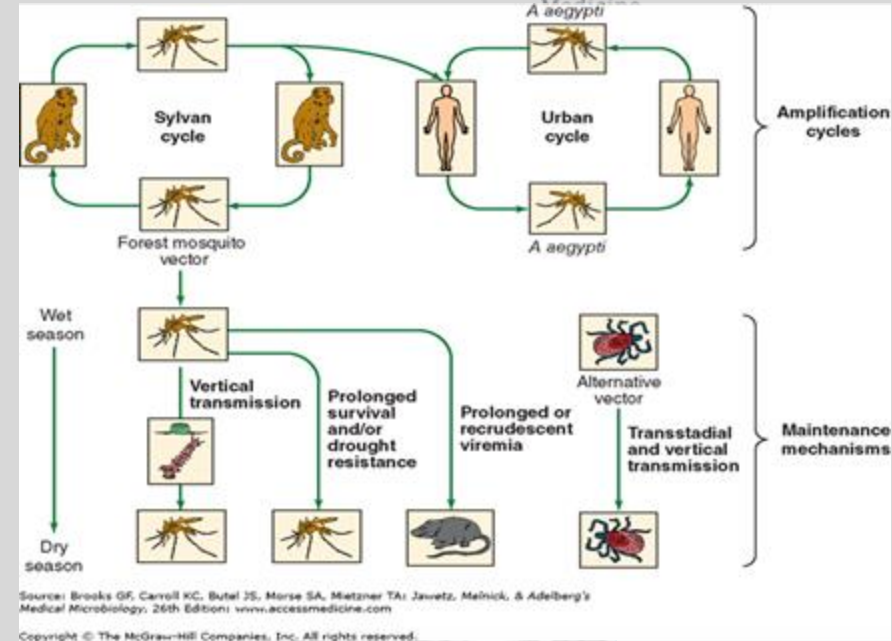


Dengue Fever Virus

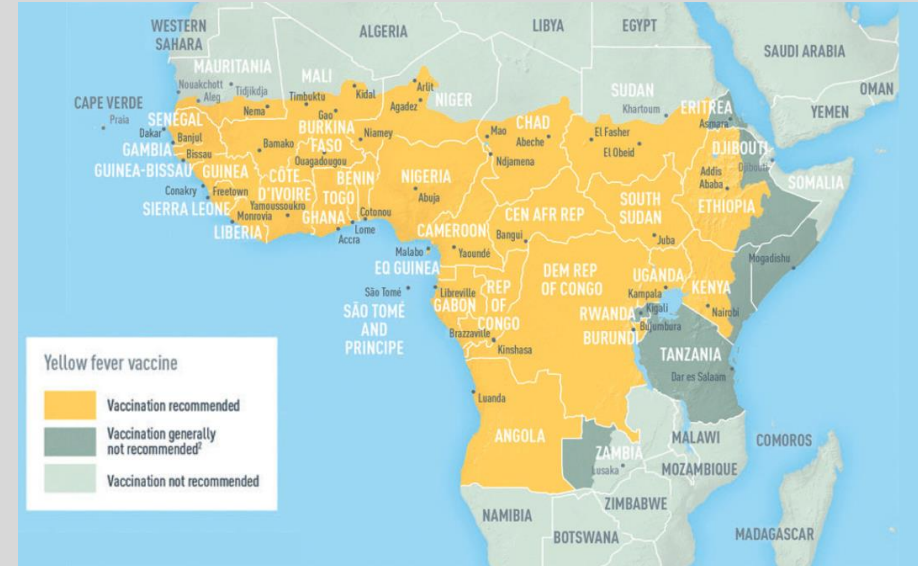
- Lab: RT-PCR
- Treatment mainly supportive
- Four viral serotypes
- Infection with one confers immunity but may cause Antibody-Dependent cytotoxicity
- 50 million cases worldwide (400k DHF)

Yellow Fever Virus

- Family: Flaviviruses
- South America & Africa
- Transmission: *Aedes aegypti* Mosquito
 - Sylvatic (jungle) Yellow Fever: Monkeys → Mosquitoes → Humans entering the forest
 - Urban Yellow Fever: Travelers from rural areas introduce virus into urban areas

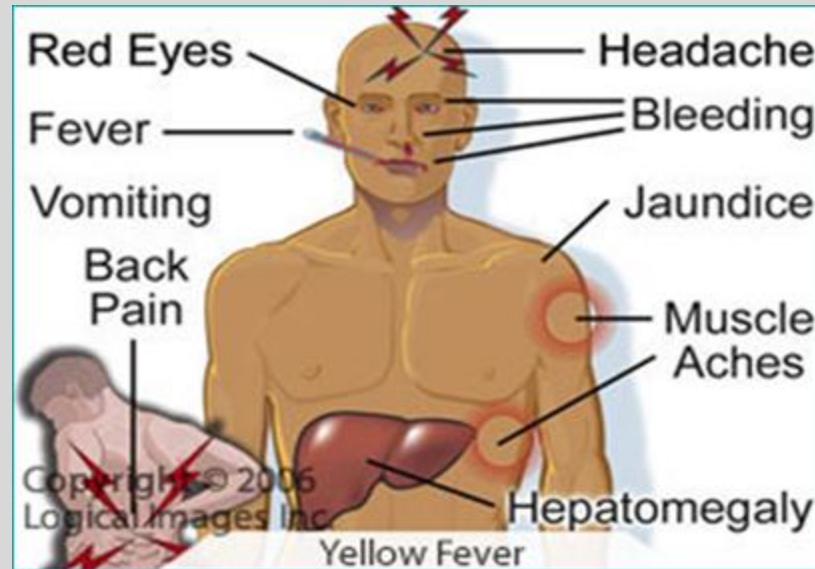


Africa



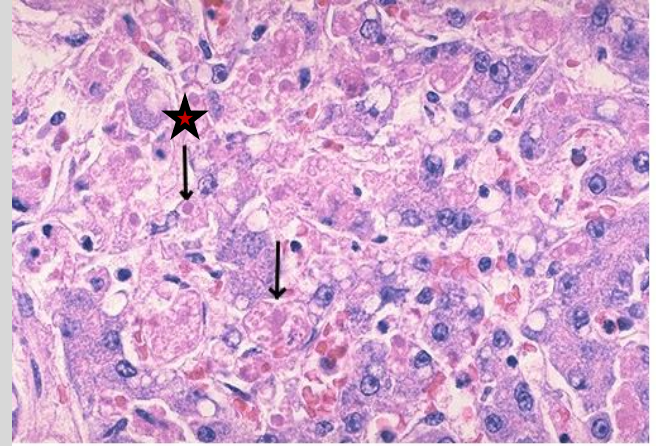
Yellow Fever Virus

Clinical features



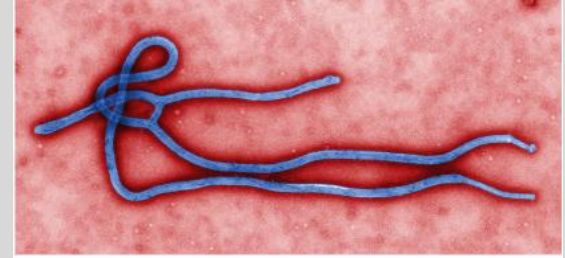
Yellow Fever Virus

- Diagnosis: ELISA/PCR
- Liver biopsy – Councilman bodies (eosinophilic apoptotic globules)
- Tx: Supportive - Most people recover within a week
- Prevention: Vaccine (live attenuated)
- Do not give to infants < 6 months, pregnant women, immunocompromised

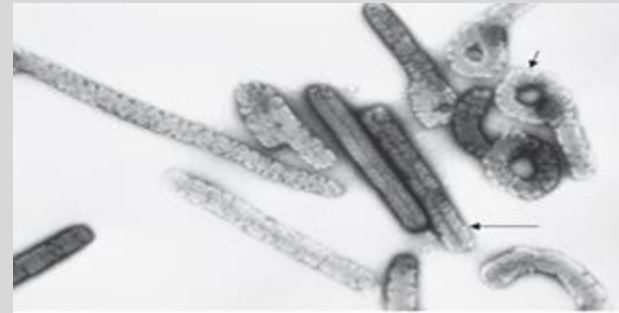


Filoviruses

- Enveloped, -ssRNA, linear
- Filoviruses replicate in multiple cell types:
 - Endothelial cells
 - Monocytes
 - Macrophage
 - Dendritic cells
- Transmission
 - African fruit bats, primates
- Ebola virus, Marburg virus



A Rousettus leschenaultii fruit bat. Kevin J. Olival/EcoHealth Alliance

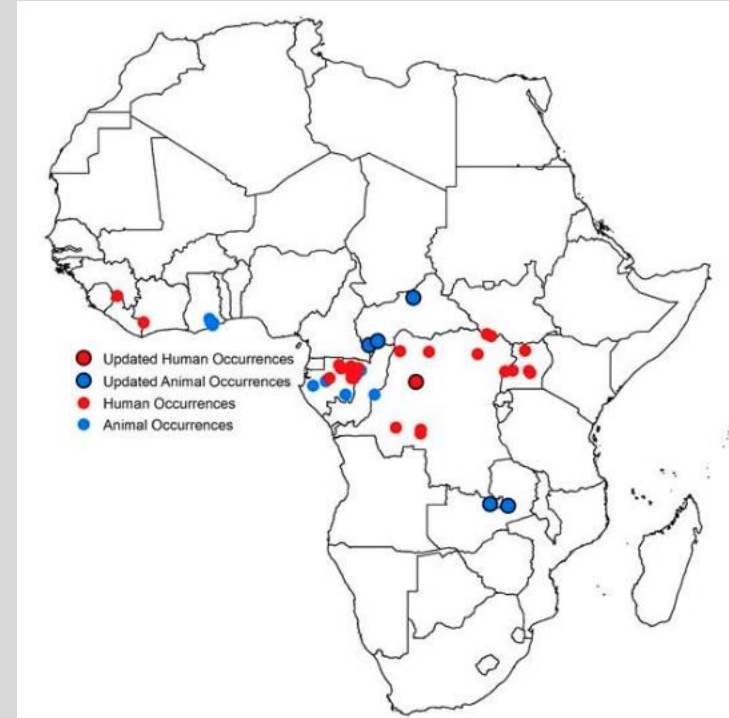


Source: Levinson W; Review of Medical Microbiology & Immunology, 12th Edition;
www.accessmedicine.com

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Ebola / Marburg

- (-) ssRNA; Filoviridae
- Central Africa
- 1st recorded cases 1967 (Marburg), 1976 (Ebola)
- Transmission from bats and primates
- Direct contact:
 - Infected blood or other bodily fluids
 - Fomites



Ebola / Marburg

Ebola / Marburg **replicates** in endothelial cells, monocytes, macrophages, and other cells.

Cytokine storm

Sepsis-like symptoms

Extensive tissue necrosis in parenchymal cells of liver, spleen, lymph nodes, and lungs

Infection of endothelial cells **impairs production of adhesion proteins**

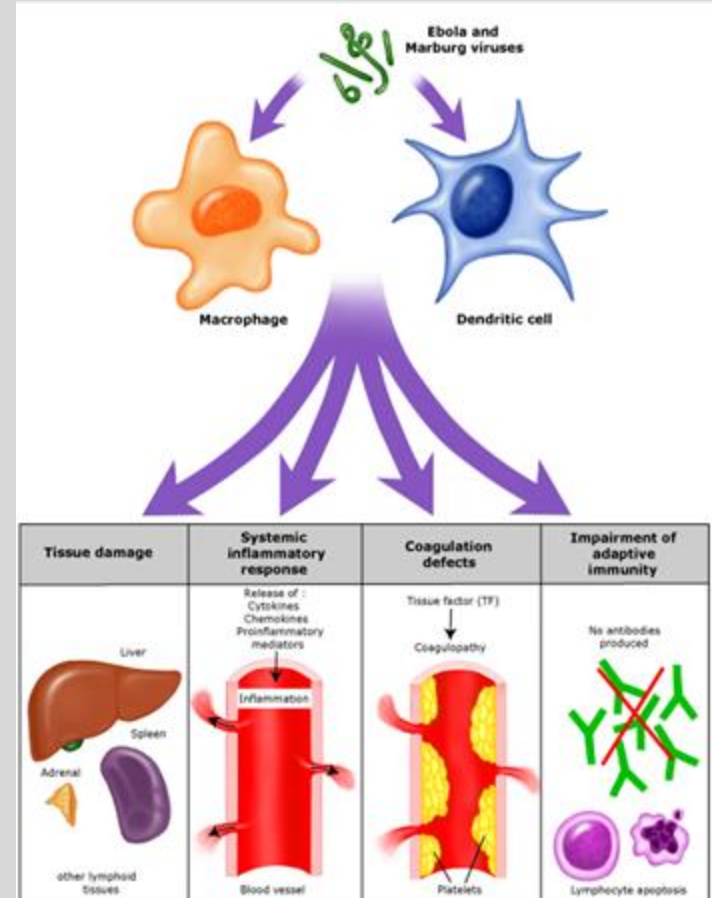
Cytolysis

Vascular injury and leakage, widespread hemorrhage

Hypovolemic shock

Disseminated intravascular coagulopathy (DIC)

Edema



Ebola / Marburg

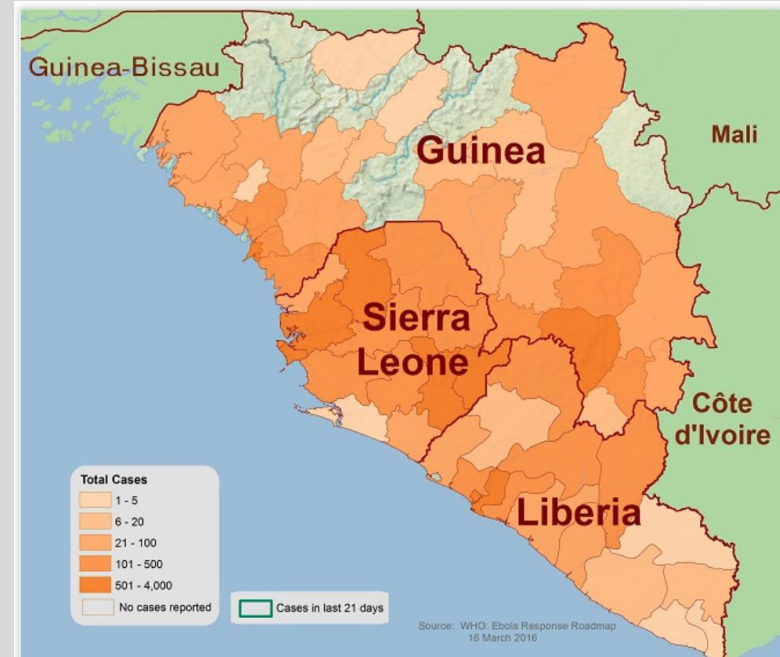
- Sudden onset of symptoms
 - Flu-like, fever, weakness, fatigue
 - Abdominal pain, nausea, vomiting, diarrhea, myalgias
 - Conjunctival hemorrhage, unexplained ecchymosis
- Symptoms increase in severity:
 - Progress to delirium, liver failure, disseminated intravascular coagulopathy, diffuse hemorrhage, shock

Ebola / Marburg

- Diagnosed with : RT-PCR
- Management: Treatment is mainly supportive
- Ebola 50–90% case fatality
- Marburg 25% case fatality
- Some of the most virulent pathogens in humans
- Ebola (Zaire, Sudan, Reston & Ivory Coast; can differ by 40%; Vaccine for Zaire)
- Ebola Virus (ERVEBO) one of six viruses said to be highly effective for particular strain but no cross protection
- First cases in Ebola in the United States (2014)



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Practice Question

A 28-year-old male presents with 5 days of temperature of 102F, severe headache, retro-orbital pain, and myalgias described as 10/10 in severity. He recently returned from a two-week vacation in Southeast Asia. Vitals are notable for a temp of 102 F, HR 100 BPM, RR 16, BP 110/70 mmHg. On physical exam, the patient is found to have a diffuse maculopapular rash and mild petechiae on the trunk and lower extremities. Laboratory results show thrombocytopenia.

1. What is the etiological agent of this disease?
2. The likely causative pathogen belongs to which viral family?
3. What is the primary mode of transmission for this infection?

References:

- Murray PR, Rosenthal KS, Pfaller MA. *Medical Microbiology*. 9th ed. Amsterdam: Elsevier; 2021.
- Centers for Disease Control and Prevention. <https://www.cdc.gov/>. Accessed March 2023. Le T, Sochat M, Schimansky S, et al. *First Aid For the USMLE Step 1*. New York: McGraw Hill; 2021.

Lecture Feedback Form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

Summary Chart

Lassa Virus	
Geography	West Africa (Endemic in Guinea, Nigeria, Liberia, Sierra Leone)
Family	Arenavirus
Transmission	Multimammate rat, <i>Mastomys natalensis</i> caught and prepared for food) Virus shed in urine & droppings • Direct contact: Touching objects, eating contaminated food, contact through cuts or sores • Aerosolized transmission • Human to human contact resulting in transmission: RARE
Clinical features	• Fever, sore throat, headache • Chest, abdominal, and back pain • Coagulopathy • Mucosal bleeding • Hemorrhage and shock 80 % of patients show no to moderate symptoms 20 % show multisystem disease 1 % of all infections result in death 15- - 20 % of hospitalizations result in death
Diagnosis	RT-PCR
Treatment	Tx: Ribavirin and Supportive Prevention: • Avoid contact with <i>Mastomys</i> rodents or infected individuals • Precautions when caring for patients with Lassa fever

Summary Chart

Hantavirus	
Geography	Global
Family	Bunyavirus
Transmission	Rodents • Aerosolized urine, droppings, saliva, dust from nests • Direct introduction into broken skin or mucous membranes
Clinical features	Hemorrhagic Fever with Renal Syndrome Leads to Acute Renal Failure • Fevers, chills • Headaches, abdominal pain, back pain • Redness of eyes • Pulmonary edema • Shock
Diagnosis	RT-PCR
Treatment	Supportive

Summary Chart

Dengue	
Geography	Southeast Asia, Western Pacific, Latin America, South America
Family	Flavivirus
Transmission	<i>Aedes mosquito</i> Incubation period 8-10 days 4 distinct, closely related serotypes for dengue
Clinical features	• Fevers, headache, rash • Myalgia, arthralgia – “Break bone” fever • Thrombocytopenia • Retro-orbital pain • Septic Shock Dengue Hemorrhagic Fever • Potentially deadly complication of dengue fever • High fever • Hemorrhagic (often with hepatomegaly) • In severe cases, circulatory failure may occur • Most serious in children • Symptoms of bleeding usually occur after 3-5 days of fever • Antibody-dependent enhancement
Diagnosis	NAAT
Management	Tx: Supportive

Summary Chart

Yellow Fever	
Geography	Africa, South America
Family	Flavivirus
Transmission	<i>Aedes mosquito</i> - vector
Clinical features	• Myalgia, predominantly back pain • Headache, loss of appetite, nausea, vomiting, epistaxis, black vomitus • Fever is paradoxically associated with slow pulse Sylvatic (jungle) Yellow Fever: Monkeys → Mosquitoes → Humans entering the forest Urban Yellow Fever: Travelers from rural areas introduce virus into urban areas Intermediate Cycle: Semi-domestic mosquitoes infect both monkey & human hosts and increased contact between man and infected mosquito leads to disease.
Diagnosis	• ELISA • Liver biopsy shows Councilman bodies (eosinophilic apoptotic globules)
Management	Tx: Supportive - Most people recover within a week Prevention: • Vaccine is live and attenuated • Do not give to infants < 6 months, pregnant women, immunocompromised

Summary Chart

Ebola / Marburg	
Geography	Central Africa
Family	Filovirus
Transmission	African fruit bats and primates Mechanism of transmission into humans unclear, but infection can be spread through direct contact with: • Infected blood or other bodily fluids • Fomites (Objects contaminated by infected bodily fluids, dead bodies) No carrier state
Clinical features	Sudden onset Sx: • Flu-like Sx, fever, weakness, fatigue • Abdominal pain, nausea, vomiting, diarrhea, myalgias • Conjunctival hemorrhage, unexplained ecchymosis Symptoms increase in severity: • Delirium, shock, liver failure, massive hemorrhaging, multiorgan dysfunction • Progress to disseminated intravascular coagulopathy, diffuse hemorrhage, shock
Diagnosis	RT-PCR
Management	• Tx: Supportive • Case fatality rate: • Ebola hemorrhagic fever: 50 % and up to 90% • Marburg hemorrhagic fever: approximately 25% • Among the most virulent pathogens of humans • Occurrence of a single human infection outside of Africa can result in an outbreak.

Summary Chart

<u>Viral Family</u>	<u>Virus</u>	<u>Disease</u>	<u>Reservoir/Vector</u>
Arenavirus	Lassa Virus	Lassa Fever	Reservoir - Rodents (Rats)
Bunyavirus	Hantavirus	Hemorrhagic Fever with Renal Syndrome	Reservoir - Rodents (Mice)
Flavivirus	Dengue Virus	Yellow Fever	Vector - Aedes Mosquito
Flavivirus	Yellow Fever Virus	Dengue Fever	Vector - Aedes Mosquito
Filovirus	Ebola Virus	Ebola Hemorrhagic Fever	Reservoir - Fruit bat or primates
Filovirus	Marburg Virus	Marburg Hemorrhagic Fever	Reservoir – Fruit bat or primates